

---

# Pooling and Drift in Delayed Bandits

---

Melika Baghi

Georgia Institute of Technology  
mbaghi3@gatech.edu

## Abstract

A system often has to act long before it learns whether the act worked: a recommender sees a click in seconds and a purchase in days. For  $K$  actions and a delay of  $d$  rounds, the sharpest published rates are minimax-optimal in the delay when only the action is visible (Zimmert and Seldin, 2020), and  $\tilde{O}(\sqrt{(K+d)T})$  over  $T$  rounds once an intermediate state is (Esposito et al., 2023). Both are governed by the number of actions, so a longer menu is always more expensive to learn from, and neither adapts to instances in which many actions are effectively interchangeable. They need not be. If the outcome depends on the action only through the state it produced, a click or a browse or a cart, then one late outcome informs every action that could have produced the state observed, and the price is set by how many genuinely different states the actions produce. We charge each delayed outcome through the state and pay an *effective dimension*  $v_t$ , between 1 and the number of states, in place of  $K$ . For the single-copy algorithm that is actually run we prove  $\tilde{O}(\sqrt{V^-} + \sqrt{dT})$  for any budget fixed in advance, and  $\tilde{O}(\sqrt{(d+1)V \log K})$  for a rotating variant; merging similar states lowers the price again, at a bias we make explicit. The gain has a limit: even when handed the exact losses from  $d$  rounds ago, no algorithm escapes  $\Omega(\sqrt{d\mathcal{E} \min\{1 + \log J, T/d\}})$ , where  $J$  counts the drifting directions and  $\mathcal{E}$  bounds how far the losses move while the learner waits. On generated data the state channel cuts regret by up to 79 per cent against action-level weighting, and on the funnel family by 32 to 68 per cent against Zimmert and Seldin (2020) under the tuning protocol we report. The guarantees assume the action-to-state map is known, and every experiment is synthetic.

## 1 Introduction

Operations commit before they learn. A markdown is set and sell-through is known a season later, staff are rostered and the shift’s outcome lands after it, marketing spend is committed before attribution resolves, an item is recommended and the purchase it may cause is days away. In each the decision is made now and the quantity it will be judged by arrives late, while something cheaper arrives quickly: a fitting-room visit, a shift’s first hour, a click. Whether that early signal can cut the cost of learning is a question between two fields. What may be inferred from the signal is a matter of data; the wait before the outcome resolves is a constraint on the operation, which no estimator removes, and the decision commits while only the first of the two is settled.

The cost of learning under delay should be set by how many genuinely different regimes the actions induce rather than by how many actions there are: a catalogue of thousands of items that funnels users through half a dozen engagement patterns should be no harder to learn from than a catalogue of six.

The mechanism that makes it true is that the outcome depends on the action only through the intermediate state it produced, so one late outcome can be charged to every action that could have produced the state observed rather than to the one action played. Bandits with intermediate observa-

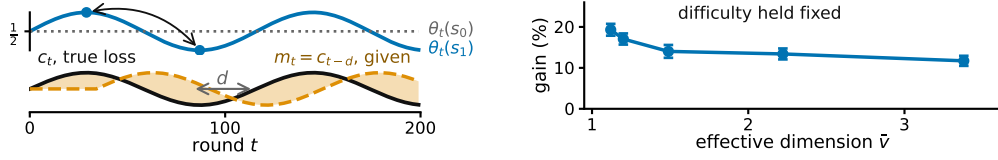


Figure 1: **The claim and its boundary, on generated data.** **Left**, what each state predicts changes while the action-to-state map stays fixed, so losses from  $d$  rounds earlier grow inaccurate;  $E_2$  of Section 2 tracks it. **Right**, more overlap allows more sharing, at matched difficulty,  $K = 40$ ,  $|\mathcal{S}| = 8$ ,  $T = 10000$ ,  $d = 50$ , 20 paired seeds, bars one standard error of the paired difference (Appendix E, study 10).

tions formalize the setting: the learner picks one action and sees the consequence of that action alone. Esposito et al. (2023) show that the *state-to-loss* map decides the complexity, an unrestricted one giving only the rate available with no intermediate observations; we fix that map and ask what the *action-to-state* structure buys. The delayed-feedback literature has since developed along other axes: Bar-On and Mansour (2025) obtain instance-dependent bounds when the observations themselves evolve, and Levy et al. (2025) treat contextual actions under delay with general function approximation. In neither does an intermediate state carry one delayed outcome across several actions.

What the pooling costs is an *effective dimension*  $v_t$ , measuring how distinguishable the action-induced state distributions are under the current play; it lies between 1 and the number of states and can fall far below  $K$ . Theorem 1 pays  $v_t$  in place of  $K$  for the algorithm that is actually run, merging similar states lowers it again at a bias we make explicit, and across the families the measured gain tracks  $v_t$  rather than  $K$ . The claim has a boundary. Waiting carries a cost of its own, which sharing cannot remove. Writing  $m_t = c_{t-d}$  for the action-loss vector from  $d$  rounds earlier and  $E_2 = \sum_t \|c_t - m_t\|_\infty^2$  for how far the losses move while the learner waits, Theorem 2 bounds below the regret this forces even when  $m_t$  is supplied free (Figure 1). Adapting to overlap and drift at once remains open (Section 5). Four ingredients separate the neighbouring settings, and it is their combination that is new here.

|                           | delayed | state seen | $P$ known | instance-dependent |
|---------------------------|---------|------------|-----------|--------------------|
| Zimmert and Seldin (2020) | ✓       | ✗          | ✗         | ✗                  |
| Esposito et al. (2023)    | ✓       | ✓          | ✗         | ✗                  |
| Eldowa et al. (2024)      | ✗       | ✓          | ✓         | ✓                  |
| Bar-On and Mansour (2025) | ✓       | ✗          | ✗         | ✓                  |
| this paper                | ✓       | ✓          | ✓         | ✓                  |

Where this work sits: the four ingredients that separate the neighbouring settings.

The last row is the contribution: no prior work combines all four. Eldowa et al. (2024) is closest, and  $v_t - 1$  is exactly the  $\chi^2$  mutual information they introduce, so that quantity is theirs; what is new is its role once the outcome is delayed (Appendix C). Vernade et al. (2020) vary the action-to-state map while fixing the state-to-loss map, where we do the reverse; Wang et al. (2026) and Zhang et al. (2026) measure a prediction error as  $E_2$  does, without the delay; and feedback graphs (Alon et al., 2015; Esposito et al., 2022; Gabbianelli et al., 2023) reveal other actions' losses, where here the state's loss must still be estimated.

## 2 Problem formulation

The question is whether the intermediate state can lower the cost of learning below what  $K$  alone forces. The model that makes it precise is the following. There are  $T$  rounds,  $K$  actions, a finite state space  $\mathcal{S}$ , a known action-to-state matrix  $P$ , and a fixed known delay  $d$ ; all logarithms are natural and  $\tilde{O}$  hides logarithmic factors. Every round follows the chain  $A_t \xrightarrow{P} S_t \xrightarrow{\theta_t} X_t$ : the learner draws  $A_t$  from a distribution  $x_t$ , observes  $S_t \sim P(\cdot | A_t)$  at once, and receives the delayed outcome only at time  $t + d$ , encoded as a loss  $X_t \in [0, 1]$  so that smaller is better, with  $\mathbb{E}[X_t | \mathcal{F}_{t-1}, A_t, S_t] = \theta_t(S_t)$ , where  $\mathcal{F}_{t-1}$  is everything seen before round  $t$ ; outcomes are conditionally independent across rounds given the states. The conditional mean depends on  $A_t$  only through  $S_t$ .

This *state-sufficiency* assumption licenses cross-action pooling: an outcome observed at state  $S_t$  may update every action that could have produced that state. If an action affects the outcome directly, in a way not represented in  $S_t$ , the argument no longer applies. Here  $x_t$  depends only on  $\mathcal{F}_{t-1}$ ;  $X_r$  may first affect the action distribution at round  $r + d + 1$ , equivalently the round- $t$  decision uses outcomes only from rounds  $r \leq t - d - 1$ .

The matrix  $P \in [0, 1]^{K \times |\mathcal{S}|}$  is row-stochastic and fixed, and the *state-loss means*  $\theta_t \in [0, 1]^{|\mathcal{S}|}$  are *oblivious*: chosen in advance, not reacting to the learner. Action losses are  $c_t = P\theta_t$  and performance is the static *pseudo-regret*, the gap to the best single action in hindsight,  $\mathbb{E}[R_T] = \mathbb{E}[\sum_t c_t(A_t)] - \min_a \sum_t c_t(a)$ . **Oracle for the lower bound.** For Theorem 2 only, the learner also receives the *stale action-loss vector*  $m_t \triangleq c_{t-d}$  before choosing  $A_t$ , with  $m_t = c_1$  for  $t \leq d$ . This is not a function of the observables, so a lower bound surviving it is stronger, and STATE-EXP3 never uses it. We measure how misleading  $m_t$  is by  $E_2 = \sum_t \|c_t - m_t\|_\infty^2 \leq T$ , which stays small when errors are small even if frequent. The geometry of  $P$  governs sharing across actions and the evolution of  $\theta_t$  governs staleness across time.

### 3 What the theory guarantees

Two results follow: the number of actions stops setting the price for the algorithm one would actually run, and no sharing removes the cost of waiting. A rotating variant and one for merged states are in Appendix B.

An arriving outcome need not update the action played alone. Every action can be credited by how likely it was to have produced the observed state. Here  $P$  is known and  $m_t$  is not given. For each state  $s \in \mathcal{S}$  let  $q_t(s) = \sum_{a=1}^K x_t(a)P(s | a)$ , and give each action  $a \in \{1, \dots, K\}$  the estimate  $\hat{c}_t(a) = P(S_t | a)X_t/q_t(S_t)$ . It is correct on average, and its noise is governed by the *effective dimension*  $v_t = \sum_s q_t(s)^{-1} \sum_a x_t(a)P(s | a)^2 \in [1, |\mathcal{S}|]$ . STATE-EXP3 runs  $m$  copies of EXP3 (Auer et al., 2002) in rotation at learning rate  $\eta$ , copy  $i \in \{0, \dots, m-1\}$  taking every  $m$ th round, and adds  $\hat{c}_t$  to a copy's running total once the outcome is usable. A round costs  $O(K)$  (Appendix A, with pseudocode and the  $\chi^2$  reading of  $v_t$ ).

**Theorem 1.** *Let  $P$  be known and  $(\theta_t)$  oblivious, and fix in advance any  $V^-$  with  $\sum_t (v_t - 1) \leq V^-$  always. Then STATE-EXP3 with  $m = 1$  and any  $\eta \leq 1/(e(d+1))$  satisfies  $\mathbb{E}[R_T] \leq \sqrt{2(3.3V^- + 2dT)} \log K + e(d+1) \log K + d$ .*

So  $K$  is replaced by  $V^-$ , which counts how differently the actions behave rather than how many there are, and the delay enters additively at  $\tilde{O}(\sqrt{V^-} + \sqrt{dT})$ , for the algorithm every experiment below runs. Writing  $\bar{v}$  for the largest  $v_t$  over all ways of playing,  $V^- = (\bar{v} - 1)T$  is admissible and depends on  $P$  alone. Rotating  $d+1$  copies and merging similar states are handled in Appendix B; none of the three adapts to  $E_2$ .

The proof turns on one pathwise identity. The denominator of  $\hat{c}_t$  is the state marginal of the distribution supplying its numerator, so  $\langle x_t, \hat{c}_t \rangle = X_t \leq 1$  on every path, whatever  $q_t(S_t)$  is. That bounds how far the play distribution moves while an outcome is in flight, which is what the delayed cross-term needs; centring by  $X_t \mathbf{1}$  replaces  $v_t$  by  $v_t - 1$  (Appendix F).

**Where this stops.** Waiting carries a cost that persists even when the stale losses are supplied free. Theorem 2 isolates that cost from time alone: the construction gives each of the  $J$  drifting directions a private state, which removes useful pooling and forces  $J \leq |\mathcal{S}| - 1$  (Appendix D).

**Theorem 2.** *Fix  $T$ , a delay  $1 \leq d \leq T/2$ , a drift budget  $\mathcal{E} \leq T/4$ , and  $1 \leq J \leq \min\{K-1, |\mathcal{S}|-1\}$ . Every algorithm, even one handed the exact losses from  $d$  rounds ago, meets some instance fixed in advance with  $E_2 \leq \mathcal{E}$  on which  $\mathbb{E}[R_T] = \Omega(\sqrt{d\mathcal{E} \min\{1 + \log J, T/d\}})$ . Without that help, and if  $K \leq T$ , the best regret any algorithm can guarantee is  $\Omega(\sqrt{KT} + \sqrt{d\mathcal{E} \min\{1 + \log J, T/d\}})$ , up to constants.*

The first term is the usual cost of exploring. The second is the cost of delay: it grows with the wait, how far the losses move during it, and the number  $J$  of directions the best fixed action can use. They come from different hard instances, so adding them overstates the truth by at most a factor of two.

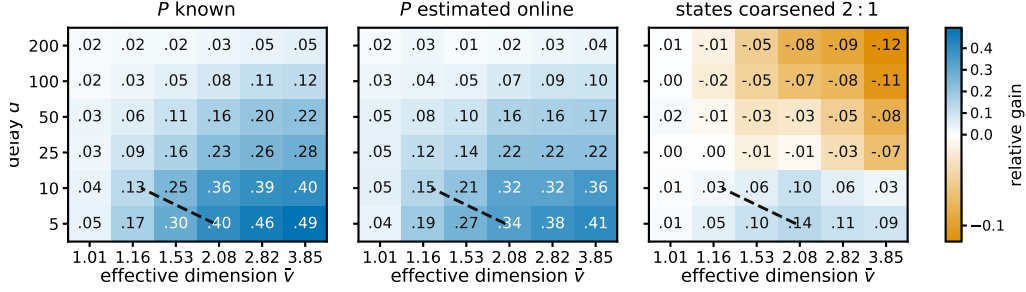


Figure 2: **Where pooling pays.** Row overlap against delay, in the three settings the studies treat separately:  $P$  known,  $P$  estimated, states coarsened. The gain should follow the effective dimension rather than the number of actions and should survive when  $P$  is estimated; both hold. Relative gain  $(R_{\text{action}} - R_{\text{state}})/R_{\text{action}}$ , single copy,  $K = 40$ ,  $|\mathcal{S}| = 8$ ,  $T = 5000$ , 12 paired seeds; blue is a gain.  $\hat{v}$  estimates  $\sup_x v(x)$  for the raw map, which in the coarsened panel is not the coarsened map’s dimension that the grouping bound charges; the dashed line compares two upper bounds for the rotating variant, so it is a reference, not a prediction for the variant plotted.

## 4 Experiments

A claim about instances is settled where the two counts it separates, how many actions there are and how many distinct behaviours they have, come apart. The families below are built so they do. With  $K = 200$  generated items and six levels of engagement the estimated dimension is 1.97, and the state-pooled variant cuts regret against action-level weighting by 79.3, 63.6 and 38.9 per cent at  $d = 10, 50, 200$ , and against the minimax-optimal delayed-bandit method of Zimmert and Seldin (2020) tuned over a grid by 67.9, 51.4 and 31.5. That setting is hand-structured, not fitted to data, so it shows the mechanism, not a deployed system.

| $d$ | action-level | tuned (Zimmert and Seldin, 2020) | STATE-EXP3 | cut   | cut   |
|-----|--------------|----------------------------------|------------|-------|-------|
| 10  | 4577         | 2959                             | 949        | 79.3% | 67.9% |
| 50  | 4752         | 3557                             | 1728       | 63.6% | 51.4% |
| 200 | 5307         | 4735                             | 3245       | 38.9% | 31.5% |

Funnel family,  $K = 200$ ,  $|\mathcal{S}| = 6$ ,  $T = 20000$ , 8 paired seeds; cuts are against action-level weighting and against the tuned baseline (study 16).

These runs use the single-copy algorithm that Theorem 1 covers; Appendix E also reports the rotating variant. A best-state rule leads seven of nine settings, but where its assumption fails its regret grows linearly, at four times our spread across seeds.

## 5 Discussion

Two things stand between an action and its worth: which actions resemble it, and how long the answer takes. The first is exploitable and checkable before deployment, since  $P$  is known and an estimate of  $\bar{v}$  well below  $K$  says whether the state channel is worth using. The second is not: Theorem 2 shows the uncertainty left by waiting survives even when the exact losses from  $d$  rounds ago are handed over free. Two limits remain:  $\bar{v}$  is estimated rather than solved for, and the bounds lie on different axes.

## References

- N. Alon, N. Cesa-Bianchi, O. Dekel, and T. Koren. Online learning with feedback graphs: beyond bandits. *COLT*, PMLR 40, 2015.
- P. Auer, N. Cesa-Bianchi, Y. Freund, and R. E. Schapire. The nonstochastic multiarmed bandit problem. *SIAM Journal on Computing*, 32(1):48–77, 2002.

- 149 Y. Bar-On and Y. Mansour. Non-stochastic bandits with evolving observations. *ALT*, PMLR 272,  
150 2025.
- 151 N. Cesa-Bianchi and G. Lugosi. *Prediction, Learning, and Games*. Cambridge University Press,  
152 2006.
- 153 N. Cesa-Bianchi, C. Gentile, and Y. Mansour. Delay and cooperation in nonstochastic bandits. *JMLR*,  
154 20, 2019.
- 155 K. Eldowa, N. Cesa-Bianchi, A. M. Metelli, and M. Restelli. Information capacity regret bounds for  
156 bandits with mediator feedback. *JMLR*, 25(353), 2024.
- 157 E. Esposito, S. Masoudian, H. Qiu, D. van der Hoeven, N. Cesa-Bianchi, and Y. Seldin. Delayed  
158 bandits: When do intermediate observations help? *ICML*, PMLR 202, 2023.
- 159 E. Esposito, F. Fusco, D. van der Hoeven, and N. Cesa-Bianchi. Learning on the edge: online  
160 learning with stochastic feedback graphs. *NeurIPS*, 35, 2022.
- 161 G. Gabbianelli, G. Neu, and M. Papini. Online learning with off-policy feedback. *ALT*, PMLR 201,  
162 2023.
- 163 O. Levy, L. Erez, A. Cohen, and Y. Mansour. Regret bounds for adversarial contextual bandits with  
164 general function approximation and delayed feedback. *arXiv:2510.09127*, 2025.
- 165 T. S. Thune, N. Cesa-Bianchi, and Y. Seldin. Nonstochastic multiarmed bandits with unrestricted  
166 delays. *NeurIPS*, 32, 2019.
- 167 C. Vernade, A. György, and T. Mann. Non-stationary delayed bandits with intermediate observations.  
168 *ICML*, PMLR 119, 2020.
- 169 S. Wang, A. Barik, and V. Y. F. Tan. Bandit convex optimization with gradient prediction adaptivity.  
170 *arXiv:2605.22191*, 2026.
- 171 K. W. Zhang, T. Baldwin-McDonald, K. Ciosek, L. Maystre, and D. Russo. Impatient bandits: opti-  
172 mizing for the long-term without delay. *JMLR*, to appear, 2026.
- 173 J. Zimmert and Y. Seldin. An optimal algorithm for adversarial bandits with arbitrary delays. *AIS-*  
174 *TATS*, PMLR 108, 2020.

## 175 A The state-pooled estimator

176 **The algorithm.** Inputs are the action-to-state matrix  $P$ , the delay  $d$ , the number of copies  $m$ , and  
177 the rate  $\eta > 0$ . The interleaved setting is  $m = d + 1$  and the single-copy one is  $m = 1$ ; Theorems 3  
178 and 1 cover them respectively. Round  $r$ 's outcome becomes usable at round  $r + d + 1$ , which for  
179  $m = d + 1$  is the next round belonging to the copy that played round  $r$ .

- 180 1. Set  $L_i \leftarrow (0, \dots, 0) \in \mathbb{R}^K$  for  $i = 0, \dots, m - 1$ , and let  $\mathcal{P}$  be an empty table of rounds  
181 awaiting their outcome.
- 182 2. For  $t = 1, \dots, T$ :
  - 183 (a) *Deliver.* If  $r \triangleq t - d - 1 \geq 1$ , read  $(i_r, q_r(S_r), S_r, X_r)$  from  $\mathcal{P}$ , remove it, and update  
184 only that copy,  $L_{i_r}(a) \leftarrow L_{i_r}(a) + P(S_r \mid a)X_r/q_r(S_r)$  for every  $a$ . This is the  
185 pooled step, and it touches every action rather than the one played.
  - 186 (b) *Select the copy.*  $i \leftarrow t \bmod m$ .
  - 187 (c) *Play.*  $x_t(a) \propto \exp(-\eta L_i(a))$ , draw  $A_t \sim x_t$ , and observe  $S_t$  at once.
  - 188 (d) *Record.* Compute the scalar  $q_t(S_t) = \sum_a x_t(a)P(S_t \mid a)$  and store  $(i, q_t(S_t), S_t, \cdot)$   
189 in  $\mathcal{P}$ , to be completed when  $X_t$  arrives at time  $t + d$ .

190 Steps (a), (c) and (d) are each  $O(K)$  and there is no optimization subproblem, so a round costs  $O(K)$   
191 once  $P$  is stored. The state is held in the  $m$  vectors  $L_i$ , which is  $mK$  numbers, the matrix  $P$ , which  
192 is  $K|S|$  numbers, and four scalars for each of the at most  $d$  rounds in flight. Only the scalar  $q_r(S_r)$   
193 is needed from round  $r$ , not the vector  $x_r$ .

194 **A worked example.** Take three actions and two states, with

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \\ 0 & 1 \end{pmatrix}, \quad x_t = (\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{4}),$$

195 so  $q_t(s_1) = \frac{1}{2}(0.8) + \frac{1}{4}(0.4) + \frac{1}{4}(0) = 0.5$  and  $q_t(s_2) = 0.5$ . The third action cannot produce  
 196  $s_1$  at all. Suppose the learner draws  $a_1$ , observes  $s_1$ , and receives  $X_t = 1$  at time  $t + d$ . Then  
 197  $\hat{c}_t(a) = P(s_1 | a)/0.5$ , so  $\hat{c}_t = (1.6, 0.8, 0)$  against the action-level  $\tilde{c}_t = (2, 0, 0)$ . One observation  
 198 updates all three actions:  $a_2$  is charged although it was never played, because it had a 0.4 chance of  
 199 producing the state seen, and  $a_3$  is charged nothing, because it could not have produced that state.  
 200 Nothing is biased by this. With  $\theta_t = (1, 0)$  the true losses are  $c_t = (0.8, 0.4, 0)$ , and averaging  $\hat{c}_t$   
 201 over  $S_t \sim q_t$  returns exactly that. What is bought is the second moment:  $v_t = 1.44$  here, against  
 202  $K = 3$  for the action-level estimate, and with  $\theta_t = (1, 1)$  both are attained exactly.

203 *Remark 1* (The  $\chi^2$  form). Under  $x_t$  the pair  $(A_t, S_t)$  has law  $x_t(a)P(s | a)$  with marginals  $x_t$  and  
 204  $q_t$ , so

$$\begin{aligned} \chi^2(P_{A_t, S_t} \parallel P_{A_t} \otimes P_{S_t}) &= \sum_{a, s} \frac{(x_t(a)P(s | a) - x_t(a)q_t(s))^2}{x_t(a)q_t(s)} \\ &= \sum_{a, s} \frac{x_t(a)P(s | a)^2}{q_t(s)} - 2 + 1 = v_t - 1, \end{aligned}$$

205 the cross term summing to  $\sum_{a, s} x_t(a)P(s | a) = 1$  and the last to  $\sum_{a, s} x_t(a)q_t(s) = 1$ . So  
 206  $v_t = 1$  exactly when  $A_t$  and  $S_t$  are independent, which pins the rows of  $P$  only on the support  
 207 of  $x_t$  and therefore means all rows agree whenever  $x_t$  has full support, as exponential weights  
 208 does. If  $P$  is deterministic and every state is reached by some action in the support of  $x_t$ , then  $v_t$   
 209 equals the number of states that support reaches. In particular, under full-support play and an onto  
 210 deterministic  $P$ ,  $v_t = |\mathcal{S}|$ . This is an identity, not a bound; it is used for reading  $v_t$ , and once in the  
 211 proof of Theorem 1.

212 **Proposition 1.** Fix  $t$ , let  $x_t$  be the play distribution,  $q_t(s) = \sum_a x_t(a)P(s | a)$  the induced state  
 213 distribution, and  $\hat{c}_t(a) = P(S_t | a)X_t/q_t(S_t)$ , which is well defined because only states with  
 214  $q_t(s) > 0$  occur. For statements involving  $1/x_t(a)$ , restrict to actions with  $x_t(a) > 0$ ; STATE-EXP3  
 215 itself has full support on every action. Then  $\hat{c}_t(a) \geq 0$ ,  $\hat{c}_t(a) \leq 1/x_t(a)$ ,  $\mathbb{E}[\hat{c}_t(a) | \mathcal{F}_{t-1}] = c_t(a)$ ,  
 216 and

$$\sum_a x_t(a)\mathbb{E}[\hat{c}_t(a)^2 | \mathcal{F}_{t-1}] \leq v_t := \sum_{s: q_t(s) > 0} \frac{\sum_a x_t(a)P(s | a)^2}{q_t(s)}, \quad 1 \leq v_t \leq |\mathcal{S}|,$$

217 states of zero probability being omitted throughout, which is the convention  $0/0 = 0$ . The action-  
 218 level  $\tilde{c}_t(a) = \mathbf{1}\{A_t = a\}X_t/x_t(a)$  has the same first three properties and weighted second moment  
 219 at most  $K$ .

220 *Proof.* Nonnegativity is immediate from  $X_t \geq 0$ . Given  $\mathcal{F}_{t-1}$  the state  $S_t$  has distribution  $q_t$ , and  
 221  $\mathbb{E}[X_t | \mathcal{F}_{t-1}, S_t = s] = \theta_t(s)$  by the model, since that conditional mean does not depend on  
 222 the action, so  $\mathbb{E}[\hat{c}_t(a) | \mathcal{F}_{t-1}] = \sum_s q_t(s)P(s | a)\theta_t(s)/q_t(s) = c_t(a)$ . For the range,  $q_t(s) \geq$   
 223  $x_t(a)P(s | a)$  for every  $a$ , so  $P(s | a)/q_t(s) \leq 1/x_t(a)$  and  $X_t \leq 1$ . For the second moment,  
 224  $X_t^2 \leq 1$  gives  $\mathbb{E}[\hat{c}_t(a)^2 | \mathcal{F}_{t-1}] \leq \sum_s P(s | a)^2/q_t(s)$ , and exchanging the sums gives  $v_t$ . Each  
 225 summand of  $v_t$  is at most 1, since  $P(s | a) \leq 1$  makes  $\sum_a x_t(a)P(s | a)^2 \leq q_t(s)$ , and at least  
 226  $q_t(s)$ , since Jensen applied to  $a \mapsto P(s | a)$  under  $x_t$  gives  $\sum_a x_t(a)P(s | a)^2 \geq q_t(s)^2$ . Summing  
 227 the two bounds gives  $1 \leq v_t \leq |\mathcal{S}|$ . The action-level claims are the same computation with the state  
 228 replaced by the played action, whose weighted second moment is  $\sum_a x_t(a)/x_t(a) = K$ .  $\square$

229 The two ends are attained. If every action induces the same state distribution then  $v_t = \sum_s q_t(s) =$   
 230 1, and if  $P$  is deterministic, each action reaching a single state, then every  $P(s | a)$  is 0 or 1 and  
 231 each summand is exactly 1, so  $v_t$  counts the states reached and equals  $|\mathcal{S}|$  when every state has an  
 232 action reaching it. Disjoint row supports give  $v_t = K$  rather than  $|\mathcal{S}|$ , since the summands over  
 233 one action's private states add to  $\sum_s P(s | a) = 1$ , so a sensor with many private states per action  
 234 resolves few of them. So  $v_t$  reads the overlap between the rows of  $P$  rather than their length, and

235  $\bar{v} = \sup_x v(x)$ , the supremum over play distributions, depends on  $P$  alone. An outcome observed  
 236 at a state updates the estimate for every action that can reach it, which is why the construction of  
 237 Section 3 gives each drifting coordinate a private state, forcing  $v_t = |\mathcal{S}|$  there. The immediately  
 238 observed pairs  $(A_t, S_t)$  identify  $P$  separately, though Theorem 3 assumes it known.

## 239 B Proofs: pooling and grouping

240 Two results are stated here rather than in the body, since Theorem 1 covers the algorithm actually  
 241 run. The first is the interleaved variant.

242 **Theorem 3.** *Let  $P$  be known and  $(\theta_t)$  oblivious. Pick before the run any fixed number  $V$  with*  
 243  *$\sum_t v_t \leq V$  always. Then STATE-EXP3 with  $m = d + 1$  and the matching learning rate satisfies*  
 244  *$\mathbb{E}[R_T] \leq \sqrt{2(d+1)V \log K}$ . Writing  $\bar{v}$  for the largest  $v_t$  over all ways of playing,  $V = \bar{v}T \leq |\mathcal{S}|T$*   
 245 *is one such choice, and it depends on  $P$  alone.*

246 The second merges states that are not worth telling apart. Group them with a map  $g$  from  $\mathcal{S}$  onto  
 247 a smaller set  $\mathcal{Z}$  and run the same algorithm on  $g(S_t)$ , where  $P^g(z | a) = \sum_{s: g(s)=z} P(s | a)$ .  
 248 Grouping never raises the dimension, but the estimate now tracks a group average, so the widest  
 249 spread of losses inside a group,  $\delta_t(g) = \max_z \max_{s, s': g(s)=g(s')=z} |\theta_t(s) - \theta_t(s')|$ , is the error  
 250 introduced.

251 **Theorem 4.** *Under the hypotheses of Theorem 3, running on  $g$  with  $\sum_t v_t(g) \leq V(g)$  gives*  
 252  *$\mathbb{E}[R_T] \leq \sqrt{2(d+1)V(g) \log K} + 2 \sum_t \delta_t(g)$ .*

253 Merging need not preserve state-sufficiency, so the grouped estimate is right not for  $c_t$  but for a stand-  
 254 in loss differing from it by at most  $\delta_t(g)$  on every action. Keeping states apart preserves differences  
 255 that matter but learns more slowly; merging shares more but pays  $2 \sum_t \delta_t(g)$ . The best grouping is  
 256 the true one: with 16 states from four hidden groups, sizes 1, 2, 4, 8, 16 give regret 1590, 831, 754,  
 257 920, 1080, lowest at four, which no algorithm was told. The map  $g$  is fixed in advance; learning it  
 258 from data remains open.

259 **Proof idea.** The pooled estimator is unbiased for every action, while Proposition 1 bounds its play-  
 260 weighted second moment by  $v_t$  rather than  $K$ . With  $m = d + 1$  interleaved copies, the outcome  
 261 generated on a round belonging to one copy is available before that copy acts again, so each copy  
 262 is an ordinary exponential-weights process with immediate feedback. Summing the  $d + 1$  potential  
 263 inequalities gives  $(d + 1) \log K/\eta$ , and the pooled second moments give  $\frac{\eta}{2} \sum_t v_t$ .

264 *Proof of Theorem 3.* Fix a copy  $i$  and let  $T_i$  be the number of rounds it owns, so  $\sum_i T_i = T$ . Round  
 265  $t$ 's outcome is usable from round  $t + d + 1$ , which with  $m = d + 1$  is exactly  $t + m$ , the copy's next  
 266 round, so within a copy the feedback is undelayed and  $L_i$  carries exactly the estimates of that copy's  
 267 earlier rounds. Exponential weights on nonnegative losses give, for any action  $a^*$  and any  $\eta > 0$ ,

$$\sum_{t \in i} (\langle x_t, \hat{c}_t \rangle - \hat{c}_t(a^*)) \leq \frac{\log K}{\eta} + \frac{\eta}{2} \sum_{t \in i} \sum_a x_t(a) \hat{c}_t(a)^2,$$

268 by the usual potential telescoping with  $e^{-z} \leq 1 - z + z^2/2$ , which needs  $z \geq 0$  and therefore  
 269  $\hat{c}_t \geq 0$ , supplied by Proposition 1. Each  $x_t$  is  $\mathcal{F}_{t-1}$ -measurable, because  $L_i$  then holds only rounds  
 270  $r \leq t - m = t - d - 1$ , so taking expectations turns the left side into  $\sum_{t \in i} (\langle x_t, c_t \rangle - c_t(a^*))$  by  
 271 unbiasedness and the quadratic term into at most  $\mathbb{E}[\sum_{t \in i} v_t]$ , again by Proposition 1. Summing over  
 272 the  $m = d + 1$  copies and using  $\sum_i \min_a \sum_{t \in i} c_t(a) \leq \min_a \sum_t c_t(a)$ , which is what lets the copies  
 273 be compared against one common action, gives  $\mathbb{E}[R_T] \leq (d + 1) \log K/\eta + \frac{\eta}{2} \mathbb{E}[\sum_t v_t]$ . If  $\sum_t v_t \leq$   
 274  $V$  surely then  $\eta = \sqrt{2(d+1) \log K/V}$  balances the two terms and gives  $\sqrt{2(d+1)V \log K}$ , and  
 275  $V = \bar{v}T$  is admissible by the definition of  $\bar{v}$ .  $\square$

276 The delay enters as a factor because the copies do not share their estimates, each paying  $\log K/\eta$   
 277 over  $T/(d + 1)$  rounds. Sharing them is exactly the  $m = 1$  algorithm of the open problem of  
 278 Section 5.

279 *Proof of Theorem 4.* Running the algorithm on  $g$  is running it on the sensor  $(\mathcal{Z}, P^g)$ . The surrogate  
 280 losses below depend on  $x_t$  and so are predictable rather than oblivious, but the proof of Theorem 3

uses obliviousness nowhere, only that each  $c_t^g$  is  $\mathcal{F}_{t-1}$ -measurable given the play, so it applies unchanged and bounds regret measured in the surrogate losses  $c_t^g(a) = \sum_z P^g(z | a) \theta_t(z)$ , where  $\bar{\theta}_t(z) = \sum_{s: g(s)=z} q_t(s) \theta_t(s) / q_t^g(z)$  is the group mean the grouped estimator is unbiased for, by the same computation as in Proposition 1 with  $S_t$  replaced by  $g(S_t)$ . Fix  $a$  and  $z$ . Both  $\{q_t(s)/q_t^g(z)\}$  and  $\{P(s | a)/P^g(z | a)\}$  are probability vectors on  $\{s : g(s) = z\}$ , so their  $\theta_t$ -averages lie in a common interval of length  $\delta_t(g)$  and differ by at most  $\delta_t(g)$ . Weighting by  $P^g(z | a)$  and summing over  $z$  gives  $|c_t^g(a) - c_t(a)| \leq \delta_t(g)$  for every  $a$ , hence  $\langle x_t, c_t \rangle - c_t(a^*) \leq \langle x_t, c_t^g \rangle - c_t^g(a^*) + 2\delta_t(g)$ . Summing over  $t$  adds  $2 \sum_t \delta_t(g)$ . Merging leaves  $V(g)$  no larger than a bound valid for the raw states: it suffices to merge two states, the general case following by induction on the merges. Writing  $P_1, P_2$  for the two columns,  $B_j = \sum_a x_t(a) P_j(a)$  and  $\lambda = B_1/(B_1 + B_2)$ , Cauchy-Schwarz gives  $(P_1 + P_2)^2 \leq \lambda^{-1} P_1^2 + (1 - \lambda)^{-1} P_2^2$  pointwise in  $a$ , so after dividing by the merged denominator the merged summand is at most the two it replaces.  $\square$

## C Relation to other inaccuracy measures

Write  $\Lambda_2 = \sum_t \min\{1, \|c_t - m_t\|_2^2\}$  for the inaccuracy measure of Bar-On and Mansour (2025). Unlike  $E_2$ , which charges only the worst action in a round,  $\Lambda_2$  grows with how many actions are predicted wrongly, and since  $\|v\|_\infty \leq \|v\|_2 \leq \sqrt{J} \|v\|_\infty$  for a vector  $v$  with  $J$  nonzero coordinates,

$$E_2 \leq \Lambda_2 \leq J E_2,$$

with both ends attained: the left when a single action is mispredicted, the right when all  $J$  are mispredicted equally. So a bound in  $E_2$  is never weaker and can be  $J$  times stronger, which is why Theorem 2 is stated in  $E_2$ . The two measures agree exactly when the prediction error is concentrated on one action.

## D Proofs: the lower bound

The measure  $E_2$  is controlled by the state-loss variation  $W$ , and no converse bound holds in general.

**Lemma 1** (Delay window).  $E_2 \leq d^2 W$  for  $W = \sum_t \|\theta_t - \theta_{t-1}\|_\infty^2$ . The ratio  $E_2/(d^2 W)$  tends to one when  $\theta$  drifts monotonically in one coordinate at a constant rate and  $T/d \rightarrow \infty$ , the first  $d$  rounds contributing  $\sum_{j < d} j^2$  rather than  $d^2$  each under the convention  $m_t = c_1$ .

In words, accumulated squared prediction error is at most the state-loss variation inflated by the square of the delay, and no better bound in  $W$  alone is available.

*Proof of Lemma 1.* Throughout write  $\Delta_j = \theta_j - \theta_{j-1}$ , set  $\theta_0 = \theta_1$ , so  $\Delta_1 = 0$ , and read  $c_r$  as  $c_1$  for  $r \leq 0$ , matching the convention  $m_t = c_1$  for  $t \leq d$  of Section 2. Then  $c_t - c_{t-d} = P \sum_{j=t-d+1}^t \Delta_j$ . Because the rows of  $P$  are probability vectors, averaging cannot increase the supremum norm, so  $\|c_t - c_{t-d}\|_\infty \leq \sum_j \|\Delta_j\|_\infty$ . By Cauchy-Schwarz applied to the  $d$  summands,  $\|c_t - c_{t-d}\|_\infty^2 \leq d \sum_{j=t-d+1}^t \|\Delta_j\|_\infty^2$ . Summing over  $t$  and using that each increment belongs to at most  $d$  windows gives  $E_2 \leq d^2 W$ . The ratio  $E_2/(d^2 W)$  approaches one when every increment has the same magnitude, sign and active coordinate, and some action reaches that coordinate deterministically, since then no cancellation occurs within a full window and  $P$  passes the increment through undiminished. Without the second condition the drift can lie in the kernel of  $P$  and the ratio can be zero.  $\square$

The construction in full. Take states  $s_0, \dots, s_q$ , a stationary deterministic map sending  $a_j$  to  $s_j$  for  $j \leq J$  and every other action to  $s_0$ , and independent Rademacher signs  $\sigma_b^{(j)}$  for  $2 \leq b \leq B = \lfloor T/h \rfloor$ . Set  $\theta_t(s_0) = \frac{1}{2}$  throughout,  $\theta_t \equiv \frac{1}{2}$  on block 1, and  $\theta_t(s_j) = \frac{1}{2} - \varepsilon \sigma_b^{(j)}$  on each later block  $b$ . The neutral first block makes  $c_1 = \frac{1}{2} \mathbf{1}$ , so the convention  $m_t = c_1$  for  $t \leq d$  of Section 2 supplies a constant over exactly the rounds that precede any usable feedback, which is what the isolation step below needs at  $b = 1$ . Rounds after the last complete block carry no drift, costing a constant factor.

**Lemma 2** (Rademacher lower tail). Let  $S_B$  be a sum of  $B \geq 32$  independent Rademacher signs. For every  $1 \leq u \leq \sqrt{B}/32$ ,  $\Pr(S_B \geq u\sqrt{B}) \geq \frac{u}{4} e^{-64u^2}$ .



325 *Proof.* Write  $S_B = 2X - B$  with  $X \sim \text{Bin}(B, \frac{1}{2})$ , so the event is  $\{X \geq B/2 + u\sqrt{B}/2\}$ ; passing  
326 to  $X$  removes the lattice gap, since  $X$  takes every integer value in  $[0, B]$  whereas  $S_B$  takes only  
327 those of the parity of  $B$ . Put  $n = \lfloor B/2 \rfloor$  and  $t = \lceil u\sqrt{B}/2 \rceil + 1$ , so  $X \geq n + t$  implies the event and  
328  $t \leq u\sqrt{B}$ . Consecutive binomial probabilities have ratio  $(B - n - j + 1)/(n + j)$ , whose distance  
329 from one is at most  $8j/B$ ; for  $j \leq B/16$  that is at most  $\frac{1}{2}$ , where  $\log(1 - x) \geq -2x$  holds, so  
330 summing over  $i \leq j$  gives  $\Pr(X = n + j) \geq \Pr(X = n)e^{-16j^2/B}$  with  $\Pr(X = n) \geq 1/(2\sqrt{B})$ .  
331 Summing over the  $t$  integers  $j \in \{t, \dots, 2t - 1\}$ , all at most  $B/16$  because  $u \leq \sqrt{B}/32$ , and using  
332  $t \geq u\sqrt{B}/2$ , gives  $\Pr(X \geq n + t) \geq te^{-64u^2}/(2\sqrt{B}) \geq \frac{u}{4}e^{-64u^2}$ .  $\square$

333 Three regimes give the maximal inequality the lower bound needs. Each  $S_B^{(j)}$  is sub-Gaussian  
334 with variance proxy  $B$ , so  $\mathbb{E} \max_j (S_B^{(j)})^+ \leq \sqrt{2B \log J} + \sqrt{B}$  throughout. For the matching  
335 lower bound, first suppose  $\log J \leq B/8$  and  $\log J \geq 128$ . Taking  $u = \sqrt{\log J}/128$ , which  
336 then lies in  $[1, \sqrt{B}/32]$ , makes  $e^{-64u^2} = J^{-1/2}$ , so  $\Pr(S_B^{(j)} \geq u\sqrt{B}) \geq \frac{u}{4}J^{-1/2}$  and inde-  
337 pendence across  $j$  gives  $\Pr(\max_j S_B^{(j)} < u\sqrt{B}) \leq e^{-u\sqrt{J}/4}$ , at most  $\frac{1}{2}$  beyond an absolute  $J$ .  
338 Hence  $\mathbb{E} \max_j (S_B^{(j)})^+ \geq \frac{1}{2}u\sqrt{B} = \Omega(\sqrt{B \log J})$ . Second, if  $\log J > B/8$  and  $B \geq 1024$ , take  
339  $u = \sqrt{B}/32$ , so  $u\sqrt{B} = B/32$  and  $e^{-64u^2} = e^{-B/16}$ , and  $J \geq e^{B/8}$  makes  $J e^{-B/16}$  diverge,  
340 giving  $\mathbb{E} \max_j (S_B^{(j)})^+ = \Omega(B)$ , which is  $\Omega(\sqrt{B \min\{1 + \log J, B\}})$  in that regime. Third, when  
341  $\log J < 128$  or  $B < 1024$  the quantity  $\sqrt{B \min\{1 + \log J, B\}}$  is  $O(\sqrt{B})$ , and a single walk already  
342 gives  $\mathbb{E}(S_B^{(1)})^+ = \frac{1}{2}\mathbb{E}|S_B| \geq \frac{1}{2}(\mathbb{E}S_B^2)^{3/2}(\mathbb{E}S_B^4)^{-1/2} \geq \frac{1}{2}\sqrt{B/3}$  by Hölder, since  $\mathbb{E}S_B^2 = B$  and  
343  $\mathbb{E}S_B^4 \leq 3B^2$ . Combining the three regimes,  $\mathbb{E} \max_j (S_B^{(j)})^+ = \Theta(\sqrt{B \min\{1 + \log J, B\}})$ . This is  
344 the route by which maxima of independent random walks fix the minimax rate for prediction with  
345 expert advice (Cesa-Bianchi and Lugosi, 2006).

346 **Proof idea.** Divide time into blocks of length  $d$  and give  $J$  actions private states whose losses are  
347 perturbed by independent Rademacher signs within each block. Every observation available when  
348 an action is chosen comes from an earlier block, and the supplied  $m_t = c_{t-d}$  depends only on  
349 earlier signs, so the learner cannot predict the signs governing its current block. Its expected loss is  
350 therefore  $T/2$ , while the best fixed action in hindsight benefits from the maximum of  $J$  independent  
351 random walks. Choosing the amplitude to make  $E_2 \leq \mathcal{E}$  gives the stated scale.

352 *Proof of Theorem 2.* Take  $h = d$  and  $\varepsilon = \frac{1}{2}\sqrt{\mathcal{E}/T}$ . Every outcome usable when choosing  $A_t$  comes  
353 from a round  $r \leq t - d - 1$ , and  $t \leq bh$  with  $h \leq d$  forces  $r < (b - 1)h$ , so  $A_t$  is independent of block  
354  $b$ 's own signs; the same computation places the stale vector  $m_t$  in blocks strictly before  $b$ , so handing  
355 it to the learner changes nothing. Every algorithm's expected loss therefore equals  $T/2$ , since its  
356 action is independent of the current block's signs and all states have mean  $\frac{1}{2}$  marginally. Its regret  
357 is therefore exactly the comparator's fluctuation advantage,  $\varepsilon d \mathbb{E}[\max_{j \leq J} (\sum_{2 \leq b \leq B} \sigma_b^{(j)})^+]$  with  
358  $B = \lfloor T/d \rfloor$ , over the  $B - 1$  randomized blocks. Per round  $\|c_t - m_t\|_\infty \leq 2\varepsilon$ , because the supremum  
359 norm charges only the largest coordinate and each coordinate moves by at most  $2\varepsilon$  at a block bound-  
360 ary; the gap is  $\varepsilon$  across the first randomized block, where  $m_t$  is still the neutral  $c_1$ , and lies in  $\{0, 2\varepsilon\}$   
361 thereafter. Hence  $E_2 \leq 4\varepsilon^2 T = \mathcal{E}$  holds deterministically. The budget must bind on the realized  
362 sequence, since a budget holding only in expectation would not constrain the instance selected by  
363 Yao's principle. By Lemma 2 and the discussion following it, the expected maximum of the positive  
364 parts of  $J$  independent Rademacher sums of length  $B - 1$  is  $\Theta(\sqrt{(B - 1) \min\{1 + \log J, B - 1\}})$ ,  
365 which covers both regimes and the case  $J = 1$ . The hypothesis  $d \leq T/2$  gives  $B \geq 2$  and hence  
366  $B - 1 \geq B/2$ , so replacing  $B - 1$  by  $B$  costs a factor at most  $\sqrt{2}$  in each branch of the minimum,  
367 including the saturated branch where the minimum is  $B - 1$  rather than  $1 + \log J$ . Substituting  
368  $B = \lfloor T/d \rfloor$  and  $\varepsilon$  therefore gives  $\Theta(\sqrt{d\mathcal{E} \min\{1 + \log J, T/d\}})$ . Rounds after the last complete  
369 block carry no drift and change the constant only. When the oracle is withheld, the  $\sqrt{KT}$  term is  
370 Esposito et al. (2023, Thm. 5.1), whose construction is stationary and therefore lies in the present  
371 model class.  $\square$

372 *Remark 2* (Where the bound is matched). Suppose  $X_t = c_t(A_t)$ . On instances whose drift is carried  
373 by a single action only one coordinate of  $c_t - m_t$  is nonzero, so  $\|c_t - m_t\|_2 = \|c_t - m_t\|_\infty$ ; on  
374 the family of Section 3 that coordinate has magnitude at most  $2\varepsilon \leq 1$ , so the truncation at one in  $\Lambda_2$

is inactive and  $\Lambda_2 = E_2$ . The bound of Bar-On and Mansour (2025) then reads  $\tilde{O}(\sqrt{KT} + dE_2)$ , whose  $\sqrt{dE_2}$  contribution Theorem 2 matches up to the logarithmic factor at  $J = 1$ , so on that family the drift term of the published upper bound is unimprovable in order. Its  $\sqrt{KT}$  term is not matched, the oracle supplying an entire stale loss vector, so the oracle-assisted problem need not retain bandit exploration complexity. Neither result provides an  $E_2$ -adaptive upper bound under noisy outcomes.

## E Numerical detail

Sixteen studies were run and the seven bearing on the claim are reported here, each with an observation, an interpretation and a limitation. They keep their original numbers, which index the released scripts, so the numbering is not contiguous; the code for all sixteen is available. Studies 1 to 13 draw  $P$  from a Dirichlet and 14 to 16 do not. All policies run on the same environment parameters and common random-number streams, so a seed fixes everything except which estimator is formed, and differences are taken within a seed before averaging. Round  $r$ 's outcome is consumed before round  $r+d+1$ , matching Section 2. Reported  $\pm$  is one standard error of the paired difference. *Sample-path pseudo-regret* means  $\sum_t c_t(A_t) - \min_a \sum_t c_t(a)$  on one realization of the learner and environment randomness, computed from the true losses rather than the sampled outcomes; averaging it over seeds estimates the expected pseudo-regret the theorems bound.

**Study 1. The bound holds and pooling helps.** With  $|\mathcal{S}| = 4$ ,  $T = 10000$ ,  $d = 50$ , drift band 0.05 and 100 seeds:

| $K$ | state, $m = d + 1$ | state, $m = 1$ | action, $m = 1$ | paired gain     | state better in |
|-----|--------------------|----------------|-----------------|-----------------|-----------------|
| 8   | 455.1              | 368.2          | 387.6           | $19.3 \pm 3.4$  | 73/100          |
| 20  | 618.3              | 493.7          | 547.5           | $53.8 \pm 5.6$  | 87/100          |
| 40  | 734.3              | 574.5          | 696.1           | $121.5 \pm 9.4$ | 99/100          |
| 80  | 780.4              | 614.8          | 784.0           | $169.3 \pm 9.9$ | 100/100         |

Study 1,  $|\mathcal{S}| = 4$ ,  $T = 10000$ ,  $d = 50$ , 100 paired seeds;  $\pm$  is one standard error.

**Observed.** For the interleaved variant, sample-path pseudo-regret stayed below the bound  $\sqrt{2(d+1)|\mathcal{S}|T \log K}$  in all 100 seeds at every  $K$ , that bound running from 2913 to 4228 against realized 455 to 780. The single-copy variant beat action-level weighting at every  $K$ , by a margin growing from 19 to 169 as  $K$  rose tenfold with  $|\mathcal{S}|$  fixed, and beat the interleaved variant throughout. Sweeping the delay at  $K = 40$  gave state-pooled 387, 569 and 731 against action-level 653, 699 and 773 at  $d = 10, 50$  and 200.

**Interpretation.** A margin that grows in  $K$  while  $|\mathcal{S}|$  is held fixed is the prediction that separates the two estimators, since only the action-level cost scales with  $K$ , so the improvement is attributable to the state channel. Interleaving looks like a price of the analysis rather than of the estimator. The narrowing of the gap as the delay grows is what an additive delay term predicts and a multiplicative one does not, which is the diagnostic evidence behind the open problem of Section 5.

**Limitation.** This is one synthetic family with a fixed drift band, and a simulation cannot establish that a bound holds, only that sample-path pseudo-regret stayed below it in the runs performed.

**Study 6. The effective dimension, not  $|\mathcal{S}|$ , governs the gain.** Holding  $|\mathcal{S}| = 8$ ,  $K = 40$ ,  $T = 5000$ ,  $d = 20$  and 24 seeds, and varying only the overlap between the rows of  $P$ , the measured mean of  $v_t$  moves while  $|\mathcal{S}|$  does not:

| row overlap | mean $v_t$ | state-pooled | action-level | paired gain      |
|-------------|------------|--------------|--------------|------------------|
| 0.00        | 2.719      | 392.7        | 599.3        | $206.6 \pm 20.0$ |
| 0.25        | 2.013      | 361.1        | 528.1        | $167.0 \pm 21.3$ |
| 0.50        | 1.532      | 306.5        | 406.5        | $100.0 \pm 10.5$ |
| 0.75        | 1.168      | 199.9        | 229.2        | $29.3 \pm 3.2$   |
| 0.95        | 1.009      | 49.2         | 50.8         | $1.6 \pm 0.5$    |

**Observed.** Both the regret and the advantage over action-level weighting fall monotonically with  $v_t$ , from a paired gain of 206.6 at  $v_t = 2.719$  to 1.6 at  $v_t = 1.009$ . Nothing about the raw state count changes across the sweep.

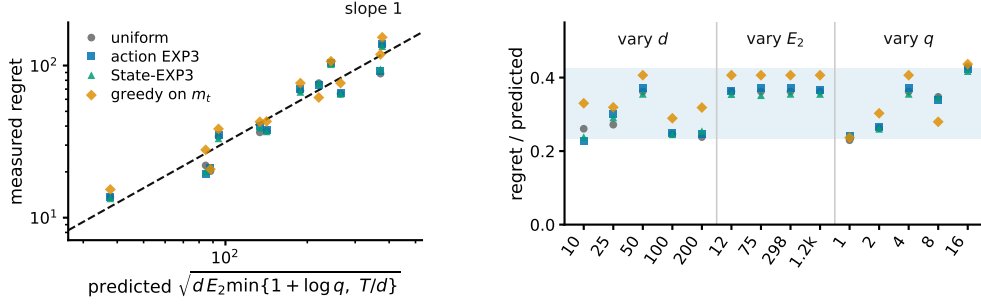


Figure 3: Study 8, the drift channel. **Left**, measured regret against the scale Theorem 2 predicts, over twelve cells in which  $d$  varies 20-fold,  $E_2$  100-fold and  $J$  16-fold. The dashed line has slope one. **Right**, the same points divided by that scale, grouped by which parameter the cell sweeps, with the shaded band spanning the observed range. STATE-EXP3 appears here only as one more algorithm on the hard family, where it has no advantage.

416 Interpretation.  $|\mathcal{S}|$  is constant and cannot explain the pattern, while  $v_t$  moves with it, which is the  
 417 comparison Theorem 3 predicts.

418 Limitation. The two estimators do not become the same object at  $v_t \approx 1$ . Rather the rows of  $P$  agree,  
 419 so every action induces the same loss and both regrets collapse, making that endpoint uninformative  
 420 rather than favorable. One parameter of one map family was varied, so the sweep is consistent with  
 421 the effective dimension governing the gain rather than establishing it.

422 **Study 8. The drift bound describes the right scaling.** The construction of Theorem 2 at  $K = 40$ ,  
 423  $T = 8000$ ,  $h = d$  and 12 paired seeds, sweeping  $d$  from 10 to 200, the amplitude  $\varepsilon$  from 0.02 to 0.20  
 424 so that  $E_2$  runs from 12 to 1193, and the drifting coordinates  $J$  from 1 to 16. We run four policies:  
 425 uniform, action-level EXP3, STATE-EXP3, and the greedy policy on the exact stale vector  $m_t$ , the  
 426 last because Theorem 2 claims to survive it.  $E_2$  is measured on the realized instance rather than  
 427 predicted, so the normalization uses the quantity the theorem uses.

428 Observed. The predicted scale  $\sqrt{d E_2 \min\{1 + \log J, T/d\}}$  ranges from 38 to 377 across the grid  
 429 and the measured regret tracks it (Figure 3). Normalized, the four policies read 0.230 to 0.429,  
 430 0.228 to 0.424, 0.238 to 0.417 and 0.237 to 0.437, a spread below 1.9 against a tenfold range in  
 431 the predictor. The oracle-assisted policy has the largest mean ratio, 0.345 against 0.311, 0.314 and  
 432 0.310 for the three that never see  $m_t$ .

433 Interpretation. The three arguments enter the predicted scale differently and the collapse holds in  
 434 all three sweeps, so it is the form of the bound rather than one fitted constant that tracks the data.  
 435 Greedy play on the exact  $m_t$  sits at the top of the band rather than below it, which is what the bound  
 436 surviving the oracle amounts to in these runs.

437 Limitation. Every cell has  $T/d > 1 + \log J$ , so only the  $1 + \log J$  branch of the minimum is  
 438 exercised and the  $T/d$  saturation branch, which needs  $d$  comparable to  $T$ , is not probed. A lower  
 439 bound quantifies over all algorithms and four policies cannot establish that. What these runs check  
 440 is the scaling of the construction.

441 **Code and environment.** The code producing every reported number and figure is at [https://](https://github.com/melikabaghi/state-exp3)  
 442 [github.com/melikabaghi/state-exp3](https://github.com/melikabaghi/state-exp3). It pins the software environment used for the reported  
 443 numbers, python 3.14.3 with numpy 2.4.2 and matplotlib 3.10.8, states the seed count for each  
 444 study, lists one command per numbered study, and commits the stored outputs each figure reads so  
 445 that every figure rebuilds without rerunning an experiment.

446 **Study 10. Overlap drives the gain, not easier tasks.** Study 6 sweeps the overlap and finds the  
 447 advantage falling as  $\hat{v}$  falls, which read at face value contradicts Theorem 3, whose bound improves  
 448 as  $\bar{v}$  falls. That sweep is confounded, since raising the overlap also flattens  $c_t = P\theta_t$  across actions,  
 449 collapsing the comparator gap so that both regrets vanish together. This study holds  $K = 40$ ,  
 450  $|\mathcal{S}| = 8$ ,  $T = 10000$ ,  $d = 50$  and the difficulty fixed, and varies only  $\hat{v}$ . Difficulty is the uniform-  
 451 play regret  $\sum_t \text{mean}_a c_t(a) - \min_a \sum_t c_t(a)$ , a property of the instance with no algorithm in it, and  
 452 the amplitude of  $\theta$  is bisected in every cell to hit a common target. The map keeps one contrast

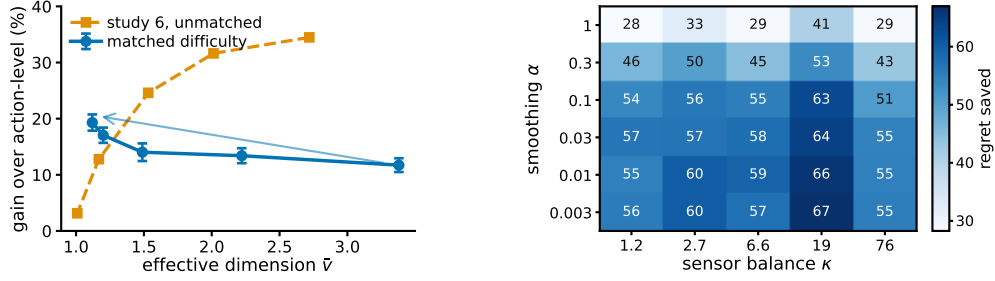


Figure 4: **Left**, study 10. The same overlap sweep with and without holding the task difficulty fixed. Matching the uniform-play regret reverses the trend, supporting the interpretation that the decline in Study 6 is driven by task-difficulty confounding rather than by the pooling mechanism itself. Bars are one standard error of the paired difference. **Right**, regret saved against action-level weighting over the smoothing pseudocount and the sensor balance, 10 paired seeds per cell, from the plug-in sensitivity study whose script is released with the rest.

453 direction alive at every overlap and  $\theta$  is aligned with it, which is what lets the gap survive as the  
 454 rows agree.

455 Observed. Across 20 paired seeds the relative gain rises from 11.7 per cent at  $\hat{v} = 3.380$  to 19.3 per  
 456 cent at  $\hat{v} = 1.120$ , with absolute gains  $85.5 \pm 8.9$  to  $106.2 \pm 7.9$  and a correlation of  $-0.929$  between  
 457  $\hat{v}$  and the gain. Study 6’s unmatched sweep falls from 34.5 to 3.1 per cent over a comparable range  
 458 (Figure 4).

459 Interpretation. The two sweeps disagree in sign, and the matched-difficulty sweep agrees with the  
 460 overlap dependence suggested by Theorem 3, although this experiment uses the single-copy variant  
 461 rather than the interleaved one that theorem analyses. Overlap is what pooling exploits, and study 6  
 462 measures overlap confounded with an instance becoming trivial.

463 Limitation. At the two highest overlaps some seeds cannot reach the common target and the realized  
 464 difficulty is 5 to 13 per cent below it, so the matching is close rather than exact. Since a smaller  
 465 gap should if anything shrink the advantage, the measured rise is conservative in that direction. The  
 466 matched gains are also smaller in magnitude than the unmatched ones, which is what fixing the  
 467 difficulty costs.

468 **Study 10b. The matched-difficulty trend survives a second configuration.** Study 10 is a single  
 469 design point, so the same procedure was re-run at  $K = 20$ ,  $|\mathcal{S}| = 12$ ,  $T = 8000$ ,  $d = 20$  and 20  
 470 paired seeds, with the common target lowered to 500 so that it is reachable at every overlap. Nothing  
 471 else changes.

| overlap | $\hat{v}$ | $R_{\text{unif}}$ | state | action | paired gain    |
|---------|-----------|-------------------|-------|--------|----------------|
| 0.00    | 3.440     | 500.0             | 364.7 | 407.2  | $42.5 \pm 8.2$ |
| 0.35    | 2.225     | 500.0             | 346.8 | 401.3  | $54.6 \pm 8.3$ |
| 0.65    | 1.518     | 500.0             | 309.3 | 372.4  | $63.2 \pm 6.4$ |
| 0.85    | 1.240     | 493.6             | 271.8 | 346.0  | $74.3 \pm 7.2$ |
| 0.95    | 1.191     | 490.1             | 256.9 | 327.6  | $70.7 \pm 7.8$ |

473 Observed. The relative gain rises from 10.4 to 21.6 per cent as  $\hat{v}$  falls from 3.440 to 1.191, and the  
 474 correlation between  $\hat{v}$  and the absolute gain is  $-0.972$  against study 10’s  $-0.929$ . Realized difficulty  
 475 spans 490.1 to 500.0, a 1.020 times spread, so the matching is tighter here than in study 10.

476 Interpretation. The direction study 10 reports is not an artifact of its single configuration. Two  
 477 independent design points, differing in  $K$ ,  $|\mathcal{S}|$ ,  $T$  and  $d$ , give the same sign and a comparable  
 478 magnitude.

479 Limitation. Both configurations use the same construction and the same matching routine, so this  
 480 tests configuration sensitivity and not construction sensitivity. Both run the practical  $m = 1$  variant.  
 481 An earlier attempt at this replication silently failed because `match_scale` binds its target as a default

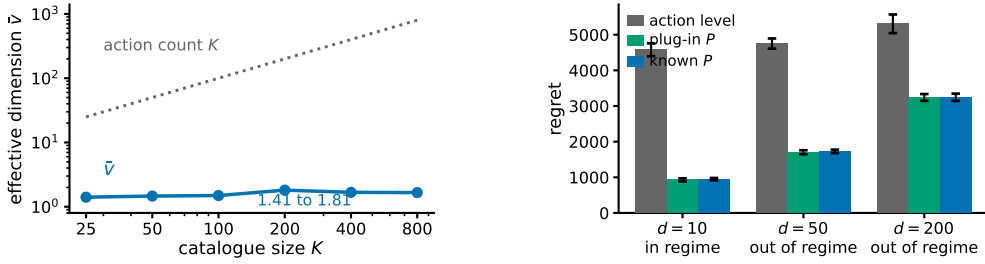


Figure 5: Study 14, a recommendation funnel. **Left**, the effective dimension stays near two while the catalogue grows thirty-two-fold, so the gap between  $K$  and  $\hat{v}$  comes from the catalogue having categories, not from a chosen  $K$ . **Right**, the three arms at  $T = 20000$  and 8 paired seeds, labelled by whether the regime condition of Section 3 holds. Bars are one standard error. No real data is used anywhere in this study.

argument, so the bisection saturated and the difficulty was never matched; the run reported here passes the target explicitly.

**Study 14. Many actions resolved by few states is a natural regime.** Every study so far draws  $P$  from a Dirichlet, which is the assumption under test rather than evidence for it. This one builds an environment to the shape of a recommendation funnel instead. Actions are  $K = 200$  catalogue items, states are  $|\mathcal{S}| = 6$  ordered engagement depths, and the outcome is a delayed conversion. Three structural facts are imposed because funnels have them, namely that items fall into a modest number of categories sharing an engagement profile, that item quality is heavy-tailed, and that the loss falls with engagement depth and drifts seasonally rather than adversarially. The generated catalogue puts 74 per cent of its mass on no engagement and 1.4 per cent on the deepest.

Observed. The effective dimension is  $\hat{v} = 1.97$  against  $K = 200$ , a hundredfold gap, and it stays between 1.41 and 1.81 as the catalogue grows from 25 to 800 items (Figure 5). Against action-level weighting the state-pooled variant saves 79.3, 63.6 and 38.9 per cent at  $d = 10, 50$  and  $200$ , the largest margins anywhere in this appendix. The plug-in matches known  $P$  at every delay, the paired difference never reaching two standard errors. The regime condition  $\hat{v}(d+1) \log K \leq K + d$  holds at  $d = 10$  and fails at  $d = 50$  and  $d = 200$ .

Interpretation. The separation between  $K$  and  $\hat{v}$  arrives from catalogue structure rather than from a tuned parameter, and it does not close as the catalogue grows, which is the claim the Dirichlet families cannot make. That the condition Theorem 3 needs fails at the delays a funnel has, while the single-copy variant still saves most of the regret, is the same message as study 1 in the setting that motivates the paper. It is evidence for the open problem of Section 5 rather than for the theorem.

Limitation. *No real data is used.* There is no interaction log in this environment, nothing here is fitted to one, and this is not a semi-synthetic benchmark in the usual sense of that phrase. The structure is imposed by hand from qualitative properties, so it shows that the regime is describable and self-consistent, not that any deployed system occupies it. The arms run  $m = 1$ , covered by Theorem 1, and the environment has no adversarial component, so it exercises the pooling channel and not the drift channel.

**Study 16. Pooling still helps against the strongest published algorithm.** Every comparison so far is against action-level weighting, EXP3-IX, naive sharing, or the best-state rule. None is the strongest published method for delayed feedback: that is the hybrid Tsallis-plus-entropy FTRL of Zimmert and Seldin (2020), whose delay term is additive rather than multiplicative. We run it on both headline families, with the same environments, seeds and delay convention, under three tunings.

| family, cell      | STATE-EXP3 | Zimmert and Seldin | same, grid-best | STATE-EXP3, grid-best |
|-------------------|------------|--------------------|-----------------|-----------------------|
| funnel, $d = 10$  | 949.0      | 2959.4             | 2413.8          | 27.0                  |
| funnel, $d = 50$  | 1728.1     | 3557.0             | 2556.4          | 95.9                  |
| funnel, $d = 200$ | 3244.8     | 4734.7             | 2807.7          | 199.7                 |
| overlap 0.00      | 661.5      | 731.8              | 610.4           | 202.4                 |
| overlap 0.65      | 584.1      | 673.1              | 544.0           | 51.2                  |
| overlap 0.95      | 461.4      | 540.2              | 419.4           | 34.3                  |

Study 16. Regret under three tunings; grid-best columns are chosen by reading these benchmarks.

Observed. With STATE-EXP3 at the single tuning used everywhere else here and Zimmert and Seldin (2020) at the best of a six-point grid around its own value, STATE-EXP3 wins every cell, cutting regret by 31.5 to 67.9 per cent on the funnel and 9.6 to 14.6 per cent on the matched sweep. With both at their grid-best, over seven multipliers for Zimmert and Seldin (2020) and six for STATE-EXP3, STATE-EXP3 wins every cell again and by more, 66.8 to 98.9 per cent. With Zimmert and Seldin (2020) at its grid-best and STATE-EXP3 left at its paper tuning, the third column against the first, Zimmert and Seldin (2020) wins the funnel at  $d = 200$  and all three matched cells, four of six.

Interpretation. No column is a symmetric protocol. The two grids are finite and unequal, and STATE-EXP3's grid-best sits at its top multiplier in every cell, so its own minimum may lie outside the range searched; the truncation runs against STATE-EXP3. Where it is not held to the worse tuning of the two, the state channel helps against the strongest published method, which is expected, since that method is action-level by construction and cannot read the state at all. Grid tuning helps STATE-EXP3 far more than it helps Zimmert and Seldin (2020), which is consistent with pooling cutting estimator variance and lower variance admitting a larger learning rate.

Limitation. The third column against the first is a real loss and is reported as one. These are our own synthetic families, and a method built for adversarial worst cases is not shown at its best on benign instances. Every grid-best multiplier is chosen by reading these benchmarks, so none of the last three columns is a procedure a practitioner could run; only the first is. Eight paired seeds on the funnel and ten on the matched sweep.

Taken together these studies cover both channels. Studies 1, 6, 10 and 14 probe the state channel and are consistent with the effective dimension, rather than  $|S|$  or  $K$ , governing what pooling buys; study 10 had to control for task difficulty before study 6's sweep pointed that way. Study 14 is the only one whose matrix is not drawn from a Dirichlet and reports the largest margins here, on a structure imposed by hand rather than measured. Study 16 places STATE-EXP3 against the strongest published method: ahead in every cell when both are tuned alike, behind in four of six when that method alone is grid-tuned.

## F Proof of Theorem 1: additive delay without interleaving

Interleaving cannot give an additive bound. Copy  $i$ , which absorbs a share  $V_i$  of the budget  $V = \sum_i V_i$ , has regret  $\sqrt{2V_i \log K}$ , and Cauchy–Schwarz over the  $m$  copies gives  $\sqrt{2mV \log K}$ , tight when the  $V_i$  agree, so the  $\sqrt{m}$  in Theorem 3 is not an artifact of the analysis. An additive bound has to come from a single copy.

Throughout,  $L_t = \sum_{r \leq t-d-1} \hat{c}_r$  and  $x_t \propto e^{-\eta L_t}$ , so  $L_{t+1} - L_t = \hat{c}_{t-d}$ , with  $\hat{c}_r = 0$  for  $r < 1$ . The filtration  $\mathcal{F}_t$  is the environment's, including outcomes not yet delivered, as in Proposition 1; under it  $x_t$  is  $\mathcal{F}_{t-d-1}$ -measurable, so  $x_{r+d}$  is  $\mathcal{F}_{r-1}$ -measurable.

**The drift lemma.** The natural route prices the change of measure from  $x_r$  to  $x_{r+d}$  through the normaliser  $Z_r$ , whose reciprocal has no uniformly bounded moment. That is avoidable. The pooled estimate satisfies

$$\langle x_t, \hat{c}_t \rangle = \sum_a x_t(a) \frac{P(S_t | a) X_t}{q_t(S_t)} = X_t \leq 1$$

identically, on every path, because the denominator is the state marginal of the same distribution that supplies the numerator.

**Lemma 3.** If  $\eta \leq 1/(e(d+1))$  then  $x_{t+1}(a) \leq (1+1/d) x_t(a)$  for every  $t$  and every  $a$ , surely, and hence  $x_{t+d}(a) \leq e x_t(a)$ .

558 *Proof.* Strong induction on  $t$ . For  $t \leq d$  nothing is delivered and  $x_{t+1} = x_t$ . For  $t > d$ ,  
 559 assume the claim for every  $s < t$  and compose it over the  $d$  steps from  $t - d$  to  $t$ , giving  
 560  $x_t(a) \leq (1 + 1/d)^d x_{t-d}(a) \leq e x_{t-d}(a)$ . Contracting with the fixed nonnegative vector  $\hat{c}_{t-d}$  and  
 561 using the identity above,  $\langle x_t, \hat{c}_{t-d} \rangle \leq e \langle x_{t-d}, \hat{c}_{t-d} \rangle = e X_{t-d} \leq e$ . With  $Z_t = \sum_b x_t(b) e^{-\eta \hat{c}_{t-d}(b)}$   
 562 and  $e^{-z} \geq 1 - z$ , this gives  $Z_t \geq 1 - \eta \langle x_t, \hat{c}_{t-d} \rangle \geq 1 - \eta e \geq d/(d + 1)$ , so  $x_{t+1}(a) =$   
 563  $x_t(a) e^{-\eta \hat{c}_{t-d}(a)} / Z_t \leq x_t(a) / Z_t \leq (1 + 1/d) x_t(a)$ .  $\square$

564 What the induction bounds is  $\langle x_t, \hat{c}_{t-d} \rangle = X_{t-d} q_t(S_{t-d}) / q_{t-d}(S_{t-d})$ , the state-marginal ratio that  
 565 the  $1/Z_r$  route could not reach. Individual  $\hat{c}_r(a)$  remain unbounded; only their  $x_t$ -weighted mass is  
 566 controlled. Cesa-Bianchi et al. (2019) prove the action-level version under  $\eta \leq 1/(Ke(d + 1))$  and  
 567 Thune et al. (2019) remove the  $K$ , using that their estimate is carried by the single action played, so  
 568 the weight cancels into a same-action ratio. A pooled estimate is spread over every action and does  
 569 not collapse that way; the identity above replaces the cancellation.

570 **Centring.** Let  $\zeta_t = \hat{c}_t - X_t \mathbf{1}$ , with  $\zeta_r = 0$  for  $r < 1$ . Exponential weights is unchanged by a  
 571 shift that is constant across actions, so  $x_t$  is the same iterate and this is a change of analysis, not of  
 572 algorithm (Eldowa et al., 2024). Keeping  $\gamma(s) = \mathbb{E}[X_t^2 \mid S_t = s]$  inside the state sum,

$$\mathbb{E}[\langle x_t, \zeta_t^2 \rangle \mid \mathcal{F}_{t-1}] = \sum_s \gamma(s) \left[ \frac{\sum_a x_t(a) P(s \mid a)^2}{q_t(s)} - q_t(s) \right] \leq v_t - 1,$$

573 each bracket being nonnegative by Jensen and  $\gamma(s) \leq 1$ , the final step being Remark 1. Bounding  
 574  $\langle x_t, \hat{c}_t^2 \rangle$  by  $v_t$  first and then subtracting does not work: it leaves  $v_t - \mathbb{E}[X_t^2]$ .

575 *Proof of Theorem 1.* Since  $\zeta_r(a) \geq -X_r \geq -1$  we have  $\eta \zeta_r(a) \geq -\eta$ , and Lagrange's re-  
 576 mainder gives  $e^{-z} \leq 1 - z + (e^\eta/2)z^2$  for  $z \geq -\eta$ . With  $\log(1 + y) \leq y$ , the potential  
 577  $W_t = \sum_a \exp(-\eta \sum_{r \leq t-d-1} \zeta_r(a))$  obeys  $\log(W_{t+1}/W_t) \leq -\eta \langle x_t, \zeta_{t-d} \rangle + (\eta^2 e^\eta/2) \langle x_t, \zeta_{t-d}^2 \rangle$ .  
 578 Telescoping from  $W_1 = K$  and using  $\log W_{T+1} \geq -\eta \sum_{r \leq T-d} \zeta_r(a^*)$  for the comparator  $a^*$ ,

$$\sum_{r=1}^{T-d} \langle x_{r+d}, \zeta_r \rangle \leq \sum_{r=1}^{T-d} \zeta_r(a^*) + \frac{\log K}{\eta} + \frac{\eta e^\eta}{2} \sum_{r=1}^{T-d} \langle x_{r+d}, \zeta_r^2 \rangle.$$

579 The shift cancels between the two sides. As  $x_r$  is  $\mathcal{F}_{r-d-1}$ -measurable,  $\mathbb{E}[\langle x_r, \hat{c}_r \rangle] = \mathbb{E}[\langle x_r, c_r \rangle]$   
 580 and  $\mathbb{E}[\hat{c}_r(a^*)] = c_r(a^*)$ , and the  $d$  rounds undelivered by  $T + 1$  contribute at most  $d$ , so

$$\mathbb{E}[R_T] \leq \frac{\log K}{\eta} + \frac{\eta e^\eta}{2} \sum_r \mathbb{E}[\langle x_{r+d}, \zeta_r^2 \rangle] + \sum_r \mathbb{E}[\langle x_r - x_{r+d}, \hat{c}_r \rangle] + d.$$

581 Lemma 3 gives  $x_{r+d} \leq e x_r$  surely, and  $x_{r+d}$  is  $\mathcal{F}_{r-1}$ -measurable, so the second sum is at most  
 582  $e \sum_r \mathbb{E}[v_r - 1] \leq eV^-$ .

583 For the third, write  $x_{r+d}(a) = x_r(a) e^{-\eta G_r(a)} / Z'$  with  $G_r = \sum_{u=r-d}^{r-1} \hat{c}_u \geq 0$  and normaliser  
 584  $Z' \leq 1$ , distinct from the one-step  $Z_t$  of the Lemma. Then  $x_r(a) - x_{r+d}(a) \leq x_r(a)(1 -$   
 585  $e^{-\eta G_r(a)}) \leq \eta x_r(a) G_r(a)$  for every  $a$ , trivially where the left side is negative. Since  $\hat{c}_r \geq 0$  and  
 586  $x_r, G_r$  are  $\mathcal{F}_{r-1}$ -measurable, the conditional expectation is at most  $\eta \langle x_r, G_r \rangle$ , and  $\mathbb{E}[\langle x_r, G_r \rangle] =$   
 587  $\sum_{u=r-d}^{r-1} \mathbb{E}[\langle x_r, \hat{c}_u \rangle] \leq d$ : for each such  $u$ ,  $x_r$  is  $\mathcal{F}_{r-d-1}$ -measurable and  $r - d - 1 \leq u - 1$ , so  
 588 it leaves the conditional expectation and  $\mathbb{E}[\langle x_r, \hat{c}_u \rangle] = \langle x_r, c_u \rangle \leq 1$ . This is measurability, not  
 589 independence, and at  $u = r - d$  the two  $\sigma$ -fields coincide, so the step rests on the timing convention  
 590 of Section 2. The sum is at most  $\eta dT$ .

591 Finally  $\eta \leq 1/(2e)$  gives  $e^{1+\eta} < 3.3$ , so  $\mathbb{E}[R_T] \leq \log K/\eta + \frac{\eta}{2}(3.3V^- + 2dT) + d$ . For  $f(\eta) =$   
 592  $A/\eta + B\eta$  restricted to  $\eta \leq \eta_{\max}$  one has  $\min f \leq 2\sqrt{AB} + A/\eta_{\max}$ , in the interior and in the  
 593 binding case alike; with  $A = \log K$ ,  $B = (3.3V^- + 2dT)/2$  and  $\eta_{\max} = 1/(e(d + 1))$  this is the  
 594 stated bound.  $\square$

595 The cap on  $\eta$  binds only when  $d \gtrsim T/(e^2 \log K)$ , and not at the settings of Appendix E: the funnel  
 596 study's own rates are 0.0058, 0.0031 and 0.0016 at  $d = 10, 50, 200$  against caps 0.0334, 0.0072 and  
 597 0.0018, so Theorem 1 covers those runs as they were executed. Over three seeds and 20000 rounds

598 the induction has room to spare,  $\max_a x_{t+1}(a)/x_t(a)$  reaching 1.006, 1.003 and 1.002 against the  
 599 permitted 1.100, 1.020 and 1.005.  
 600 The two bounds are numerically close on the funnel, where  $\bar{v} \approx 2$ : the additive form carries  $3.3(\bar{v} -$   
 601  $1) + 2d$  and the interleaved one  $(d + 1)\bar{v}$ , and these cross near  $\bar{v} = 2$ . The additive form gains as  $\bar{v}$   
 602 rises toward  $|S|$ , and it describes the algorithm that is actually run, which is the reason to prefer it.